

ELC 5365 Introduction to Deep Learning Class Project Proposal

Deep Learning for Multi-class Alzheimer's MRI Classification

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Motivation

Alzheimer's disease is an important and challenging topic in medical research because it affects memory and cognitive function, and early detection can be very helpful for treatment and care. I am interested in this topic because it connects deep learning with a real healthcare problem. MRI images contain a large amount of structural information, and deep learning provides a useful way to learn patterns from these images automatically. For me, this project is a good opportunity to apply what I learned in class, such as CNNs, model comparison, and transfer learning, to a meaningful image classification problem.

Objective

The main goal of this project is to build and compare deep learning models for multi-class classification of Alzheimer's MRI images. More specifically, I want to classify MRI images into four categories and see how well different models perform on this task. I also want to compare a basic CNN with a transfer learning model to understand whether using a pretrained network can improve performance. In addition to overall accuracy, I plan to look at confusion matrices and other metrics such as precision, recall, and F1-score, so that I can better understand where the model performs well and where it still makes mistakes.

Dataset

For this project, I plan to use the public Kaggle Alzheimer's multiclass MRI dataset. I chose this dataset because it is easy to access, organized for image classification, and suitable for training deep learning models in Colab.

The Alzheimer's Disease Multiclass Dataset was sourced from Kaggle and contains approximately 44,000 MRI images, categorized into four distinct classes based on the severity of Alzheimer's disease. This dataset supports training and testing machine learning models for automated AD classification. It is organized into four directories, each representing a different severity level:

- Non-Demented (12,800 images): Subjects with no signs of dementia.
- Very Mild Demented (11,200 images): Subjects exhibiting very mild symptoms.
- Mild Demented (10,000 images): Subjects diagnosed with mild dementia.
- Moderate Demented (10,000 images): Subjects with moderate dementia.

Method

My plan is to start with a simple CNN as a baseline model. This will help me understand the basic performance of a model trained from scratch on this dataset. After that, I will try a deeper architecture or a transfer learning model, such as ResNet, to see whether a pretrained model can achieve better results. Before training, the images will be resized and normalized, and the dataset will be split into training, validation, and test sets. During training, I will try to use techniques such as dropout, early stopping, and data augmentation to reduce overfitting. Finally, I will compare the models using accuracy, precision, recall, F1-score, and confusion matrix analysis. My goal is not only to get a good result, but also to understand how model choice and training strategy affect performance on a medical image classification task.